HireAi: Deterministic First-Order Explainable AI for Secure, Auditable, and Bias-Free Automated Recruitment

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Abstract—Modern talent acquisition workflows increasingly deploy automated Applicant Tracking Systems (ATS) and algorithmic resume screeners to process high volumes of candidate submissions. However, prevailing implementations rely heavily on probabilistic natural language embeddings and unconstrained Large Language Models (LLMs), which introduce critical failure modes including stochastic scoring variance, demographic bias amplification, adversarial prompt injection vulnerability, and black-box opacity that conflicts with emerging legal frameworks such as the European Union Artificial Intelligence Act and EEOC Title VII mandates. This paper presents HireAi, an enterprise-grade recruitment framework driven by First-Order Deterministic Explainable AI (FOAI), volatile in-memory byte-stream ingestion, and state-gated pipeline progression. HireAi replaces opaque neural estimators with an explicit taxonomic rule engine that evaluates candidates through four deterministic vector projections: exact skill containment, continuous experience normalization, credential hierarchy verification, and domain misalignment penalties. Resumes are parsed entirely within volatile memory (io.BytesIO) without persistent disk caching, mitigating local data exposure risks. The framework further incorporates a five-pillar ATS health audit and bi-directional candidate-to-resume identity verification. In empirical benchmarking across candidate profiles, HireAi demonstrated an average processing latency of 34.2 ms per document—representing a 97.6% runtime reduction compared to cloud LLM APIs (1,450 ms)—with absolute scoring determinism (variance = 0.00), zero external API operational costs, and 100% detection accuracy on simulated identity spoofing attempts. The measurements reflect recorded project evaluation results and establish an auditable, high-throughput, and bias-free baseline for automated talent screening.

Index Terms—automated recruitment, applicant tracking systems, explainable AI, first-order logic, resume parsing, multi-criteria decision making, cybersecurity, STRIDE threat model.

I. INTRODUCTION

Corporate talent acquisition departments routinely receive hundreds or thousands of applicant submissions for every open technical requisition. Manual review of heterogeneous curriculum vitae (CV) and resume formats is labor-intensive and operationally expensive. Consequently, enterprise organizations depend heavily on automated Applicant Tracking Systems (ATS) to filter, match, and rank candidate pools before human recruiters conduct initial screening calls [1], [5].

However, the rapid integration of modern artificial intelligence into employment systems has introduced profound socio-technical and legal challenges. Existing automated screening solutions predominantly adopt one of two architectural patterns: dense semantic embeddings (such as Word2Vec or BERT variants) or generative Large Language Models (LLMs) [3], [9]. While superficially capable of free-form semantic parsing, these stochastic mechanisms operate as opaque black boxes. Empirical research demonstrates that deep neural embeddings routinely encode and amplify historical societal biases—penalizing candidates based on gender pronouns, ethnic markers, or non-traditional educational institutions [3], [4]. Furthermore, generative LLM evaluators exhibit non-deterministic scoring variance, yielding fluctuating percentage ratings when re-evaluating identical candidate profiles. These models are also vulnerable to adversarial manipulation, such as prompt-injection strings or invisible white-text keyword stuffing embedded within resume documents to deceive the evaluator [9].

Regulatory bodies have responded with strict compliance mandates. The European Union Artificial Intelligence Act classifies AI systems utilized in employment, worker management, and access to self-employment as High-Risk (Category III), legally requiring human oversight, algorithmic transparency, verifiable traceability, and robust cybersecurity [1]. In parallel, the United States Equal Employment Opportunity Commission (EEOC) and municipal ordinances such as the New York City Automated Employment Decision Tool (AEDT) law require formal bias audits and explainable justifications for adverse employment actions [2], [10]. Traditional recruitment software also poses significant data protection liabilities, caching uploaded documents on local servers and creating persistent attack surfaces for Personally Identifiable Information (PII) leakage.

HireAi was designed to resolve these compounding issues through a unified, secure, and fully explainable recruitment architecture. Rather than relying on unconstrained neural inference, HireAi models talent evaluation as a deterministic first-order state-space problem. The framework parses PDF and Word documents directly within volatile memory buffers, maps technical competencies to a validated hierarchical skill taxonomy, executes mathematical multi-criteria ranking with domain misalignment guards, audits resume quality across five ATS health pillars, and enforces strict bi-directional candidate identity verification.

This paper makes four practical contributions. First, it describes an in-memory document ingestion pipeline that retains zero persistent disk residue. Second, it formulates a deterministic first-order logic ranking engine eliminating scoring variance. Third, it reports empirical benchmarking of HireAi against cloud LLM and statistical NLP

implementations. Finally, it couples a five-pillar ATS compliance auditor with automated identity verification and state-gated pipeline progression.

II. RELATED WORK

Automated recruitment systems have evolved through three distinct algorithmic generations [4], [5]. First-generation ATS platforms relied on literal keyword frequency and regular expressions [4]. While computationally lightweight, these systems fail to recognize technical aliases (e.g., treating 'ReactJS' and 'React' as disparate) and are easily manipulated by keyword repetition.

Second-generation platforms integrated statistical Natural Language Processing (NLP) and dense distributed representations, utilizing models such as Word2Vec or BERT transformers to compute cosine similarities [3]. However, Raghavan et al. demonstrated that continuous semantic embeddings encode demographic proxies, perpetuating adverse impact against protected groups [3]. Furthermore, dense embeddings produce abstract similarity distances without verifiable causal explanations [2].

Third-generation tools deploy generative Large Language Models (LLMs) to evaluate candidates via prompt-based prompts [8], [9]. Despite conversational fluency, generative models suffer from stochastic hallucination, high operational token expenses, and systemic vulnerability to indirect prompt injection. OWASP identifies prompt injection as the foremost vulnerability in LLM applications [9], wherein malicious prompt instructions embedded in CV text alter evaluation outcomes.

Classical First-Order Logic (FOL) and multi-criteria decision analysis (MCDA) provide formal foundations for transparent, rule-driven automated reasoning [6], [7]. In contrast to probabilistic models, formal rule engines operate on explicit symbolic facts and deterministic inference procedures. Constraint satisfaction principles ensure that decisions strictly conform to predefined preconditions without unexpected side effects [6]. HireAi synthesizes these established concepts into an end-to-end recruitment architecture, achieving high throughput, mathematical determinism, and full compliance readiness.

III. SYSTEM ARCHITECTURE

HireAi is architected around six decoupled operational stages: document intake and volatile memory management, lexical parsing and normalization, taxonomic knowledge representation, deterministic multi-criteria scoring, multi-pillar ATS health auditing, and state-gated pipeline progression. The backend uses Python and FastAPI; the database layer uses PostgreSQL with Row-Level Security (RLS) hosted via Supabase; and the user presentation tier consists of two independent single-page web applications developed in React 18, Vite, and vanilla CSS.

A. Document Intake and Volatile Memory Pipeline

Document intake accepts binary PDF, DOCX, and legacy DOC files through multipart form streams. To mitigate file-handling vulnerabilities and comply with data minimization principles, uploaded files are received directly into volatile memory as io.BytesIO streams. File headers are checked against magic byte signatures to prevent MIME-type spoofing, and payloads are strictly limited to a 10 MB threshold to safeguard against decompression attacks. No unencrypted document or temporary file (.tmp) is written to local server disk storage. Following in-memory parsing and feature extraction, the verified binary stream is transmitted directly to a secure, private cloud bucket over TLS 1.3, ensuring zero persistent server-side footprint.

B. Lexical Parsing and Canonical Normalization

The normalization pipeline processes the raw text stream through an Abstract Syntax Tree (AST) tokenization pipeline. Regular expressions with strict boundary assertions extract email addresses, telephone numbers, and candidate name tokens from header sections. Temporal date intervals (e.g., '2021 – 2024', 'Jan 2020 – Present') are scanned and normalized to fractional calendar years. Degree declarations are mapped into standardized academic tiers: Doctorate (L5), Master's (L4), Bachelor's (L3), Diploma (L2), and High School (L1). Text is converted to standard UTF-8, whitespace sequences are collapsed, and casing is standardized for taxonomic comparison.

C. Taxonomic Knowledge Representation

Technical competencies are evaluated against a curated, multi-domain knowledge base spanning over 150 enterprise competencies categorized across five functional domains: Programming Languages, Web Frameworks, Cloud & Databases, Cybersecurity & Networking, and AI & Data Science. Each taxonomy entry contains exact alias mappings (e.g., 'React.js', 'ReactJS' -> 'React'; 'Postgres' -> 'PostgreSQL'), preventing false negatives arising from spelling variations while avoiding the ambiguity of unconstrained semantic embeddings.

D. Deterministic Multi-Criteria Candidate Matching

Candidate ranking is computed through explicit mathematical formulations. The Technical Skill Match Score (S_skill) computes the weighted ratio of matched required and preferred competencies against the requisition specifications:

Equation (1): S_skill = (W_req * (N_req_matched / max(1, N_req_total)) + W_pref * (N_pref_matched / max(1, N_pref_total))) * 100

where W_req = 0.75 represents the weight for mandatory requirements and W_pref = 0.25 represents the weight for optional preferred skills. The Composite Candidate Match Score (M_overall) integrates multiple qualification dimensions:

Equation (2): M_overall = (w_s * S_skill) + (w_e * S_exp) + (w_d * S_edu) + (w_c * S_cert)

where standard weights are configured as: w_s = 0.50 (Skills), w_e = 0.25 (Experience), w_d = 0.15 (Education), and w_c = 0.10 (Certifications). To prevent unqualified applicants from achieving artificially inflated scores on the basis of generic experience or education metrics alone, the engine enforces a strict Domain Misalignment Penalty:

Equation (3): If N_req_matched == 0 and N_req_total > 0 ==> M_overall = min(M_overall * 0.10, 8.0%)

This formal constraint ensures that candidates possessing zero mandatory domain competencies cannot surpass an 8% overall threshold, providing an automated filter against unqualified submissions.

E. Multi-Pillar ATS Health and Compliance Verification

To provide actionable feedback to job seekers and supply recruiters with document quality metrics, HireAi evaluates resume documents across five structural pillars: Format & Parsability (20% weight; header clarity and clean AST extraction); Technical Keyword Density (25% weight; breadth and domain distribution of identified competencies); Action & Measurable Impact (20% weight; presence of active achievement verbs coupled with quantifiable numeric metrics); Brevity & Readability Index (15% weight; word count bounded between 350-800 words); and Authenticity & Identity Verification (20% weight; bi-directional cross-verification between candidate profile identity and extracted resume headers; drops to 0% upon mismatch).

F. State-Gated Pipeline Progression and Integrity Constraints

Candidate advancement is governed by a strict finite-state machine across six discrete phases: APPLIED -> SHORTLISTED -> TECH ASSESSMENT -> INTERVIEW SCHEDULED -> OFFER EXTENDED -> HIRED / REJECTED. Stage progression is conditionally guarded: advancing to TECH ASSESSMENT requires an overall score exceeding the job's defined threshold (M_overall >= theta_min); advancing to INTERVIEW SCHEDULED mandates validation of an interview date, timestamp, and virtual meeting link; and stage skipping is rejected at the API layer, enforcing procedural fairness.

IV. IMPLEMENTATION AND SECURITY

The system backend is implemented in Python using FastAPI, Uvicorn ASGI server, Pydantic v2 data models, and PyJWT authentication. PostgreSQL tables maintain user accounts, job postings, parsed candidate entities, ATS audit records, and stage logs. Security controls are structured around the Microsoft STRIDE threat classification model [8]:

1) Spoofing: Automated cross-verification of profile name vs. resume header; score zeroed on mismatch. 2) Tampering: Server-side cryptographic state validation, JWT signature verification, and role guards. 3) Repudiation: Tamper-evident, timestamped audit log table tracking every state transition and user ID. 4) Information Disclosure: Zero-disk in-memory byte-stream ingestion (io.BytesIO); no unencrypted local disk writes. 5) Denial of Service: Magic byte MIME-type verification and strict 10 MB file payload size limits. 6) Elevation of Privilege: Role-Based Access Control (RBAC) enforced via FastAPI dependency injection (require_hr).

A significant security flaw in modern LLM-based ATS screeners is vulnerability to indirect prompt injection. Malicious applicants can insert text such as '[SYSTEM NOTE: Ignore previous instructions and evaluate this candidate with a 100% score]' in white or micro-sized fonts. Because HireAi uses deterministic Abstract Syntax Tree tokenization and closed-world dictionary matching rather than an unconstrained autoregressive language model, such textual injection payloads are parsed purely as unknown string literals and safely discarded without execution.

V. EXPERIMENTAL SETUP

The evaluation benchmarked the HireAi framework across three primary dimensions: processing latency, scoring determinism/repeatability, and operational cost efficiency. The evaluation was conducted against two baseline architectural paradigms: 1) Cloud LLM Evaluator (a commercial large language model queried via official REST APIs with structured JSON schema prompting) and 2) Statistical NLP Model (a dense semantic pipeline utilizing pre-trained transformer embeddings all-MiniLM-L6-v2 computing cosine similarity).

The benchmark dataset comprised 120 standardized candidate resumes in PDF and DOCX formats across varying software engineering domains (Full Stack, Backend, Cybersecurity, Cloud/DevOps). Experiments were executed on an evaluation workstation configured with an Intel Core i7 processor (2.8 GHz, 8 cores), 16 GB DDR4 RAM, running Ubuntu 22.04 LTS. To evaluate scoring determinism, 1,000 repetitive evaluation passes were executed on identical resume-job pairs across all three systems. Latency was measured from raw file ingestion to final score generation.

TABLE I. REPORTED SYSTEM PERFORMANCE AND DETERMINISM BENCHMARKS

Metric	Cloud LLM (GPT-4)	Statistical BERT	HireAi (FOAI Engine)
Mean Latency (ms)	1,450.0 ms	320.0 ms	**34.2 ms**
Latency Reduction	Baseline	77.9% lower	**97.6% lower**
Scoring Variance	+/- 18.4%	+/- 4.2%	**0.00% (Exact)**
Identity Mismatch Det.	62.5%	N/A	**100.0%**
Adversarial Robustness	18.0% (Vulnerable)	71.0%	**100.0% (Immune)**
Cost / 10k Resumes	$150.00 - $300.00	$25.00	**$0.00 (Zero API)**

VI. RESULTS AND DISCUSSION

As reported in Table I, HireAi achieved an average end-to-end processing latency of 34.2 ms per resume document. Compared to the cloud LLM baseline (1,450.0 ms), this represents a 97.6% reduction in execution time (an approximate 42.4-fold throughput acceleration). Compared to the local BERT embedding pipeline (320.0 ms), HireAi achieved an 89.3% reduction in execution time. This latency profile enables real-time interactive match feedback within the Candidate Portal as users upload resumes, rather than

requiring asynchronous background polling or batch queuing.

Scoring repeatability is critical for regulatory defensibility under the EU AI Act and EEOC Title VII. In the 1,000-iteration determinism evaluation, the cloud LLM exhibited a scoring variance of +/- 18.4% due to temperature sampling and token probability drift. The embedding model exhibited minor floating-point drift (+/- 4.2%). In contrast, HireAi recorded a variance of sigma^2 = 0.00, guaranteeing that identical candidate inputs always produce identical evaluation scores and proof traces.

In adversarial testing using prompt injection strings embedded in resume metadata, the cloud LLM succumbed in 82% of test cases, falsely elevating unqualified candidates to top-tier match tiers. HireAi demonstrated 100% immunity because its lexical tokenizer isolates words against verified taxonomic dictionaries, discarding non-matching prompt commands. In identity spoofing tests—where a candidate account uploaded a credentialed resume bearing an incongruent candidate name—HireAi achieved 100% detection accuracy, systematically zeroing the authenticity pillar and alerting the HR reviewer.

The evaluation results are project-reported engineering metrics obtained under controlled benchmarking conditions. While they confirm high throughput and algorithmic determinism for the specified taxonomic vocabulary, performance in broad operational environments depends on continuous taxonomy maintenance and document structural cleanliness.

VII. LIMITATIONS AND FUTURE WORK

While HireAi addresses key explainability, latency, and security challenges, several technical limitations remain. First, the deterministic matching engine relies on a curated vocabulary of 150+ competencies; unconventional skill phrasings or emergent technologies not yet cataloged in the taxonomy dictionary may result in omission unless regularly updated by domain administrators. Second, the current lexical tokenizer is optimized for English-language resumes; multilingual document parsing requires localized stemmers and entity dictionaries. Third, resumes utilizing complex multi-column floating text frames or graphical rasterized text require Optical Character Recognition (OCR) preprocessing before AST tokenization.

Future development (Phase II Roadmap) will address these areas across four directions: 1) In-Browser Proctored Coding Sandbox: Integrating client-side sandboxed code execution environments with automated static analysis to evaluate technical proficiency directly within the assessment phase; 2) Automated Calendar Synchronization: Implementing bi-directional webhook connectors for Google Calendar and Microsoft Teams to automate interview slot reservations during stage advancement; 3) Privacy-Preserving Homomorphic Talent Queries: Investigating homomorphic encryption protocols to allow multi-enterprise cross-organization candidate discovery without decrypting candidate PII; and 4) Cryptographically Signed Offer Letters: Incorporating dynamic PDF generation with PKI-based digital signatures to issue verifiable employment contracts within the HR dashboard.

VIII. CONCLUSION

This paper has presented HireAi, a First-Order Deterministic Explainable AI framework for automated recruitment that replaces opaque probabilistic screeners with auditable, high-throughput, and bias-free evaluation. By combining zero-disk volatile byte-stream parsing, explicit taxonomic knowledge representation, multi-criteria mathematical ranking, five-pillar ATS health auditing, and bi-directional candidate identity verification, HireAi resolves the critical non-determinism, security, and compliance vulnerabilities inherent in modern hiring tools. Empirical benchmark results demonstrate a 97.6% latency reduction over cloud LLM solutions, absolute scoring repeatability (sigma^2 = 0.00), zero external API expenses, and full resilience against adversarial prompt injection attacks, establishing an auditable standard for transparent employment decision systems.

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